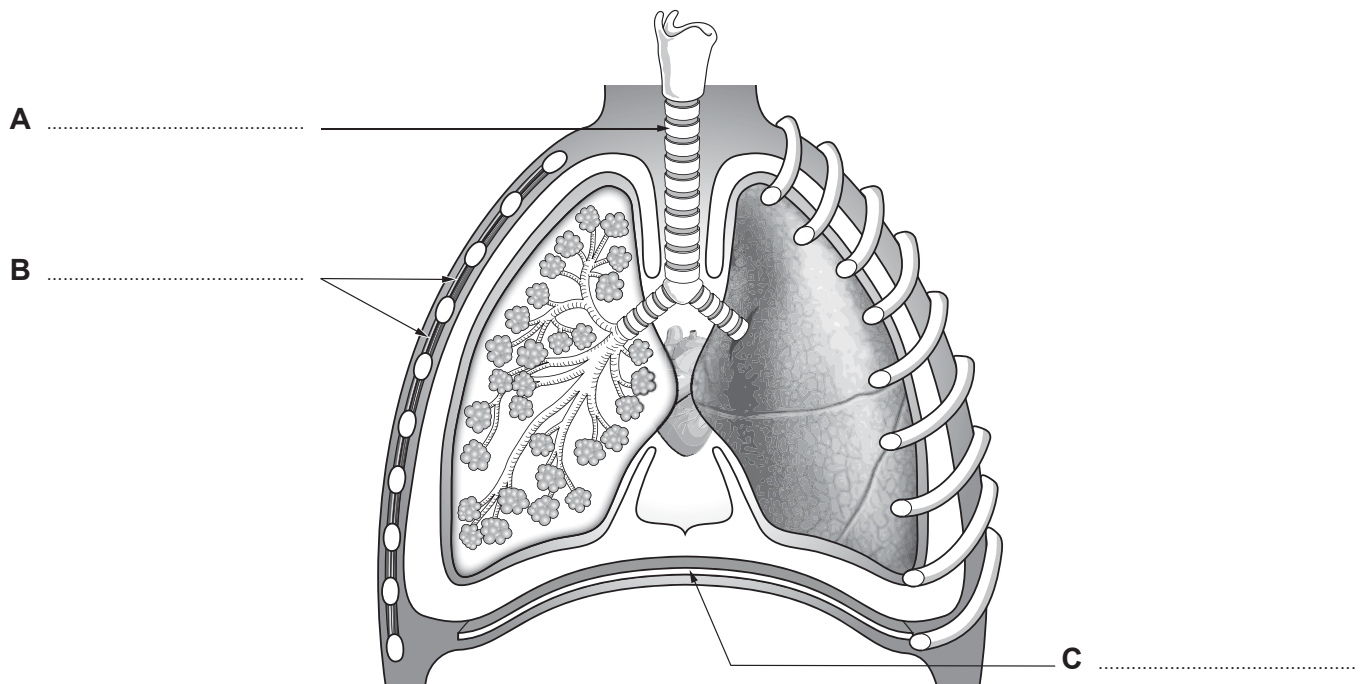


2. Sports science students are learning about the human respiratory system.



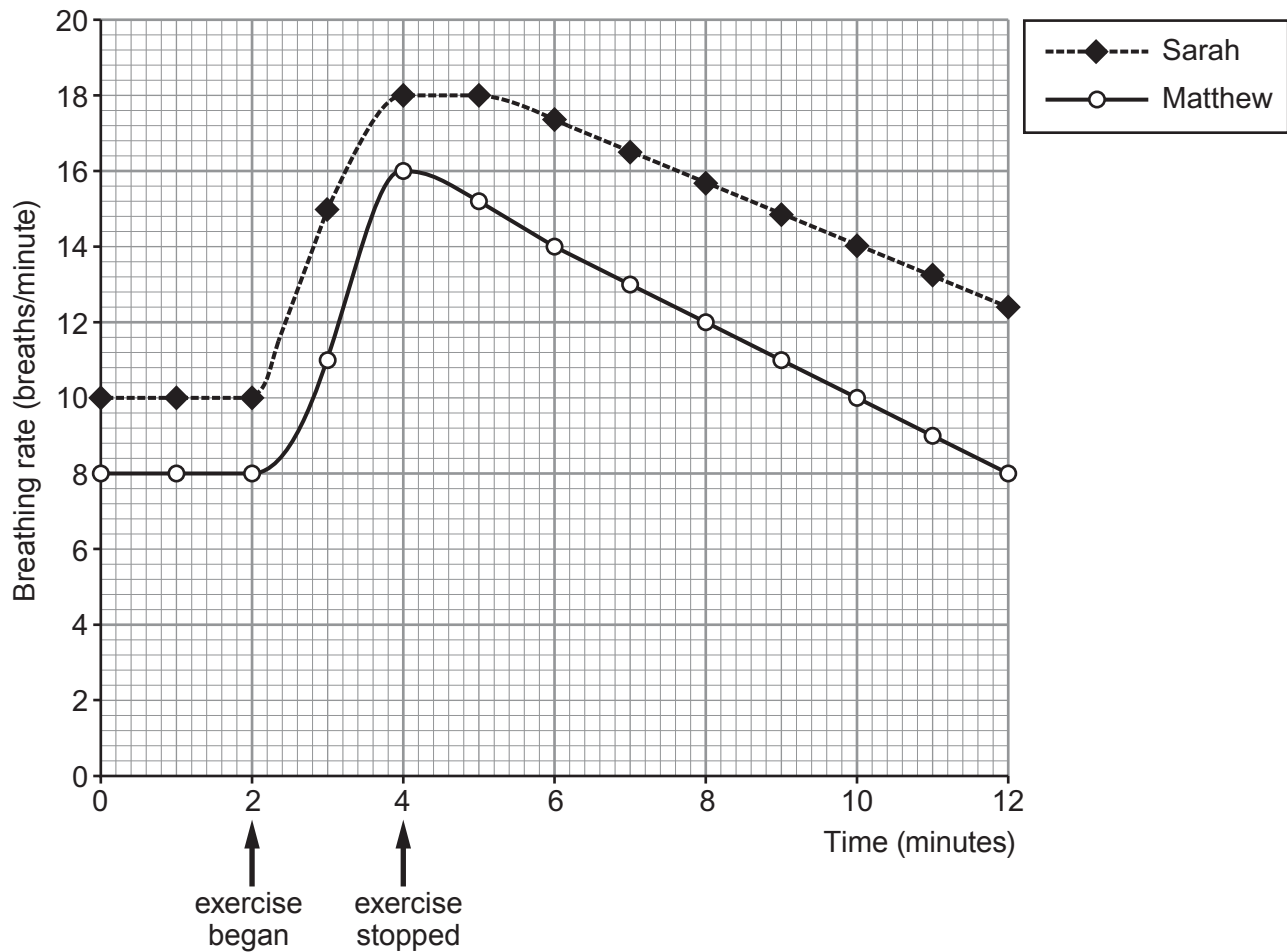
- (a) (i) Label the diagram. [3]
(ii) Explain how changes to **B** and **C** allow us to breathe in. [3]

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- (b) Two students, Matthew and Sarah, studied the effect of exercise on breathing rate. They followed the method below:

1. record breathing rate before exercising
2. exercise for 2 minutes by running on a treadmill
3. measure the breathing rates during and after exercise up until 12 minutes

Their results are shown on the graph below.



Compare the changes in breathing rate of Sarah and Matthew between 8 and 12 minutes. [3]

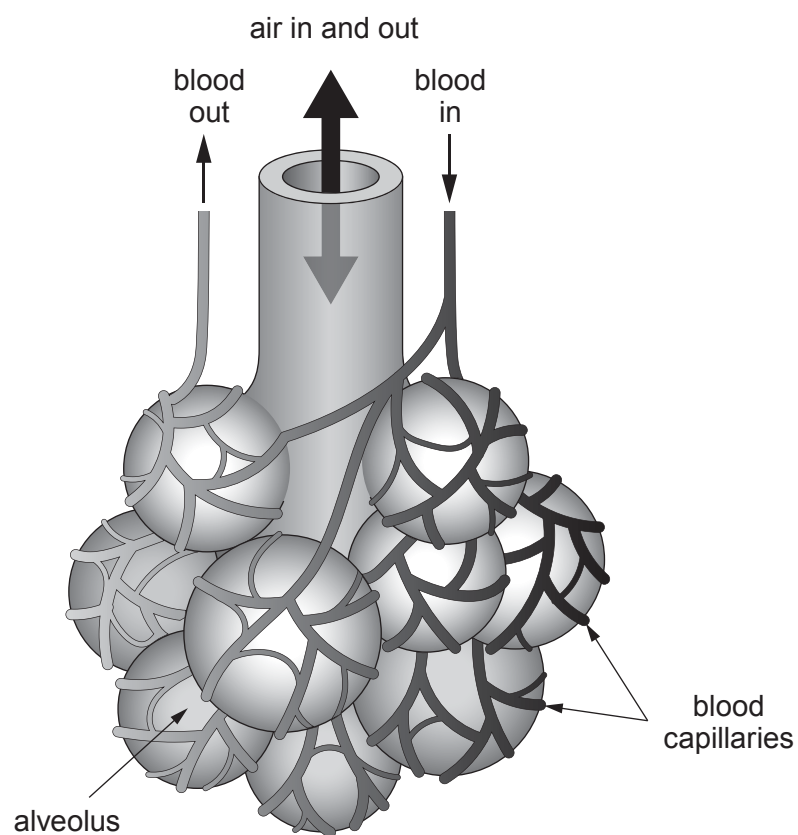
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- (c) Explain the process of gas exchange between the blood and the air inside the alveoli.

[3]



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